

Data Wrangling 🤠

Introduction to Machine Learning

Quick Poll

Where do you see AI or ML in your real life (i.e. the computer makes a decision for you)? Consider email inbox, weather, Spotify, Netflix, ChatGPT, etc.

What *can* computers do?

- Computers are **not** *smart*!
- When we write programs, **we** have to write down every rule or step:

```
>>> year = "2026"
>>> year + 1
Traceback (most recent call last):
  File "<python-input-1>", line 1, in <module>
    year + 1
    ~~~~~^~~
TypeError: can only concatenate str (not "int") to str
>>> int(year) + 1
2027
```

Weather Prediction

- Based on today's weather, how can we write a program to predict tomorrow's weather?

```
def predict_weather(today_temp, today_cloudy):  
    if today_temp > 50 and not today_cloudy:  
        print("It will be warm and sunny tomorrow.")  
    elif today_temp < 50 and today_cloudy:  
        print("It will be cool and rainy tomorrow")  
    ...  
    else:  
        print("It will be cold and sunny tomorrow.")
```

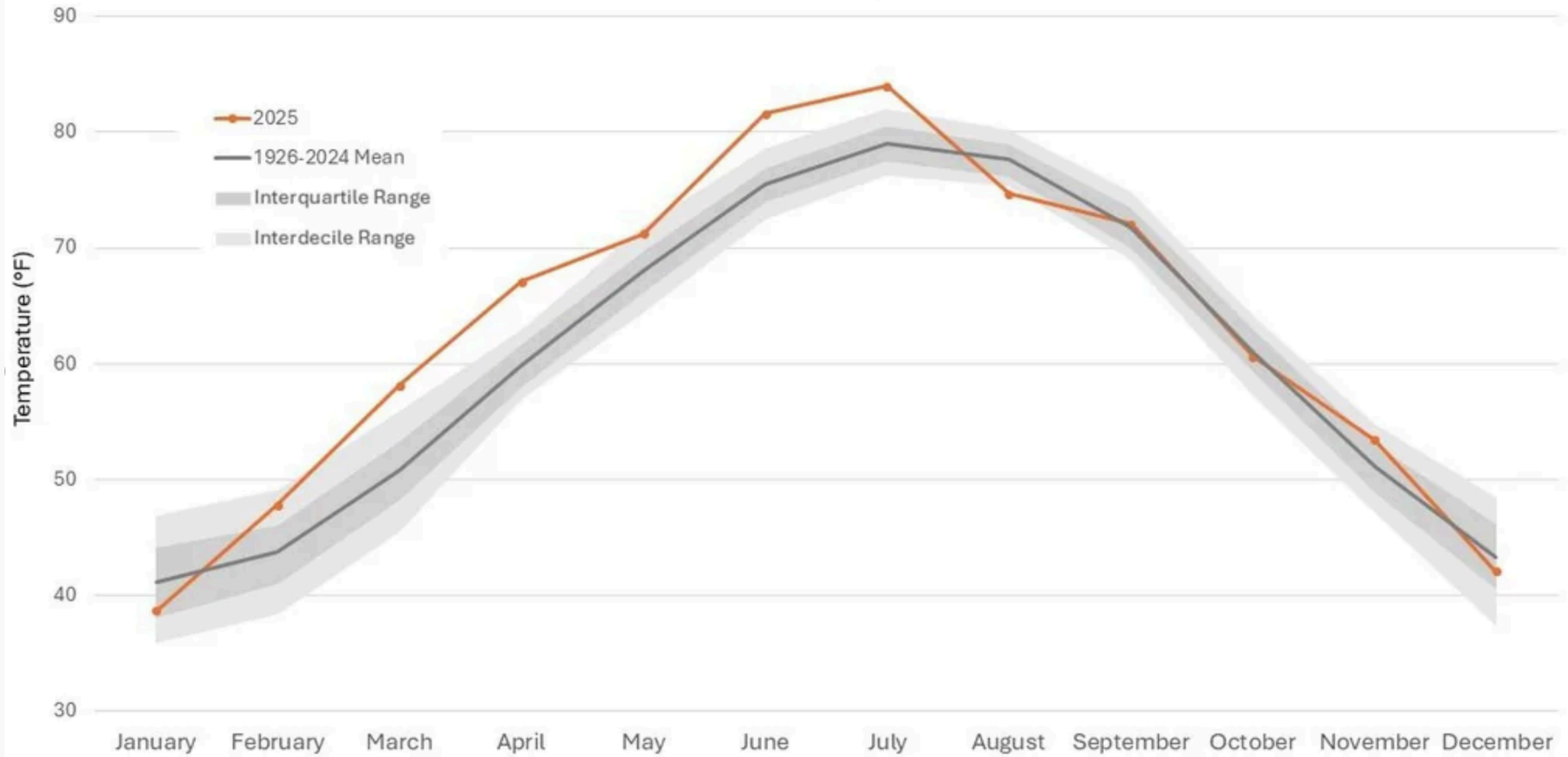
- This won't be a very good prediction! Many more factors.

Weather Prediction cont'd

- Even if we could consider more variables, how would we describe the relationships between those variables?
- It is hard to enumerate the outcome for every combination of different variables.

```
def predict_weather(today_temp, today_cloudy, today_humidity, today_air_pressure):  
    if ...  
        ...  
    elif ...  
        ...  
    elif ...  
        ...  
    else ...
```

Raleigh, NC Monthly Average Temperature



What is Machine Learning?

- Some problems are too complicated for hand-written rules
- **Data from the past** can show us interesting patterns or trends use it to **predict what may happen in the future!**
- **Machine learning** (ML) is the field where computers learn from data of the past to make decisions
 - Needs *lots of information* to learn something interesting -- ML depends on **very large** datasets!

Humans vs. Computers

- **Before*: Human programmers write down all the rules, computers follow them
- **Now with ML techniques**: Human programs will show the computers **lots of examples** and computers will find the pattern to come up with rules
- A **model** is just the pattern the computer learned and you use it to make new predictions

Training and Testing

- ML models (computer programs) depend on a **two step process**:
 - **Training**: Show your program *many* data examples from the past so it can *learn* the pattern
 - **Testing**: Show your program *new* data examples that it has never seen before and see how it *performs*

How to split up data?

- We can't grade a model on the examples it already memorized the answers to!
- So we split our labeled data into two piles:
 - **Training data** is ~80% of the original data
 - **Test data** is ~20% is of the original data
- Test data is **reserved** data, it used *only* to check how well the model does on examples it hasn't seen

Types of Machine Learning

- Two types of ML depend which on **what the contents of the training and testing datasets are**:
 - **Supervised***: A human goes through and labels all the data (i.e. someone decides which reviews are positive and which reviews are negative)
 - **Unsupervised***: The data has no labels at all so the computer learns about the data in a different way

Types of Supervised Learning

- Two types of Supervised Learning depend on **what the labels for data are**:
 - **Classification**: the label is a **category** from a fixed set (e.g. *spam / not spam, positive / negative*)
 - **Regression**: the label is a **number** (e.g. *tomorrow's high temperature, a house's price*)

Concept Check

Our capstone will predict a category based on some text (*positive* or *negative*). Does it solve a classification or regression problem?

"As with many of Wongs films, a lot of people find them to be boring and confusing. Well I like them and I like this film too. I went out and rented it on DVD and I watched it 3 times it is a very subtle movie that provides an intoxicating experience. For those who did not enjoy it, you just wasted 2 hours of your life. Too bad! Muhahahahaha!"

Concept Check

For each example, is it a **classification** or **regression**?

- Predicting a student's exam score from hours studied 
- Predicting whether an email is spam 
- Sorting photos into "cat," "dog," or "neither" 
- Predicting how many minutes a bus is delayed 

ML follows the Data Workflow!

1. **Ask** a question: Is this review positive or negative?
2. **Collect** labeled examples: find reviews humans already tagged
3. **Clean** the dataset
4. **Train**: let the computer find patterns that separate the labels
5. **Predict**: use those pattern to label *new, unlabeled* examples

Natural Language Processing

- Recall: **text** coming from human language is **unstructured data**
 - You can't put text or a transcript into a spreadsheet
- **Natural Language Processing** (NLP) is the field of study that teaches computers how to work with a process human language
 - Examples: search engines, translation apps, ChatGPT
- NLP models do not *understand* language, they find statistical patterns in language

Capstone: Naive Bayes Text Classifier

- We will combine ML + NLP to build a **text classifier**!
 - Given **textual content**, we want to assign it a **label**
 - Our model will learn labels for many different text examples and then use that knowledge to classify unseen pieces of text

Bag of Words

- How can computers can do math on text and words to learn patterns?
 - Turn words into something they can count!
- **Bag of words:** ignore word *order* and only list **which words appear**
- `"the cat sat on the mat"`
- `"the cat sat on the mat"`
- Both sentences contain the exact same set of unique words:
`{the, cat, sat, on, mat}`

Words are Evidence!

- What kinds of reviews would the word **wonderful** show up in?
- What kinds of reviews would the word **boring** show up in?
- What kinds of reviews would the word **movie** show up in?
- A classifier that counts *which* words tend to show up with which label

Model Accuracy

- Once a model makes predictions on the **test** set, how do we know if it's any good?
- **Accuracy** = number of correct predictions ÷ total predictions

```
correct = positive, predicted = positive  ✓  
correct = negative, predicted = positive  ✗  
correct = negative, predicted = negative  ✓
```

```
performance: 2 / 3 correct  ->  accuracy = 0.67
```

Concept Check

A classifier is tested on 40 reviews and gets 34 of them right. What is its **accuracy**?

Pitfalls of ML

- Will a model trained only on 10 movie reviews perform well?
- We need lots of good examples to find reliable patterns
- Note that input data can be messy, mislabeled, or exhibit bias

Summary

- In ML, instead of writing rules by hand, the computer learns them from **labeled examples**
- We split data into **training** and **test** sets so we can honestly check how well a model generalizes the things it learns
- Our text classifier will take in some **training data** with **textual content** and **labels** and learn associations between given words and labels
- The **bag of words** put every unique word into a bag...